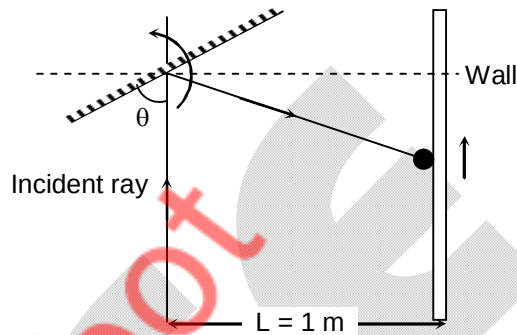


12. A droplet of water in uniform fog is formed. It then falls sweeping up the fog lying in its path. Assume that it retains all the fog in its path, remains spherical and experiences zero drag. Drop attains a constant value of acceleration after a large time interval. Its acceleration is given by $\frac{g}{K}$. Find the value of K.
13. A light ray is incident on a rotating mirror, which is rotating with angular speed 3 rad/s as shown in figure. The speed of bright spot on the wall when $\theta = 30^\circ$.

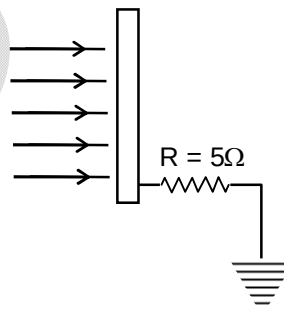


SECTION – C (Numerical Answer Type)

This section contains **TWO (02) paragraphs**. Based on **each paragraph**, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Paragraph for Question Nos. 14 and 15

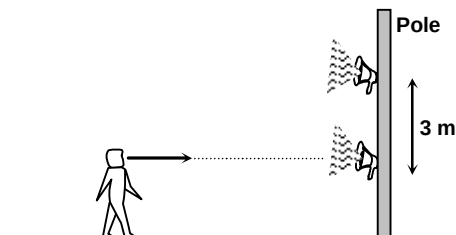
A large metal plate having area 1 m^2 is exposed to light of intensity 6.62 W/m^2 . Frequency of light is $3.2 \times 10^{15} \text{ Hz}$ which is greater than threshold value of photoemission. Emitted electrons are swept away immediately after emission. ($h = 6.62 \times 10^{-34}$)



14. The current through resistor in steady state in ampere is.....
15. The potential of plate in steady state in volt is

Paragraph for Question Nos. 16 and 17

An oscillation of frequency 680 Hz drives two speakers. The speakers are fixed on a vertical pole at distance 3 m from each other as shown in the figure. A person whose height is almost same as that of the lower speaker walked towards the lower speaker in a direction perpendicular to the pole. Assuming that there is no reflection of sound from the ground and speed of sound is $v = 340 \text{ m/s}$. Answer the following questions.



16. As the person walks towards the pole, his distance from the pole when he first hears a minimum in sound intensity is nearly
17. As the person walks towards the pole, the total number of times the person hears a minimum in sound intensity will be